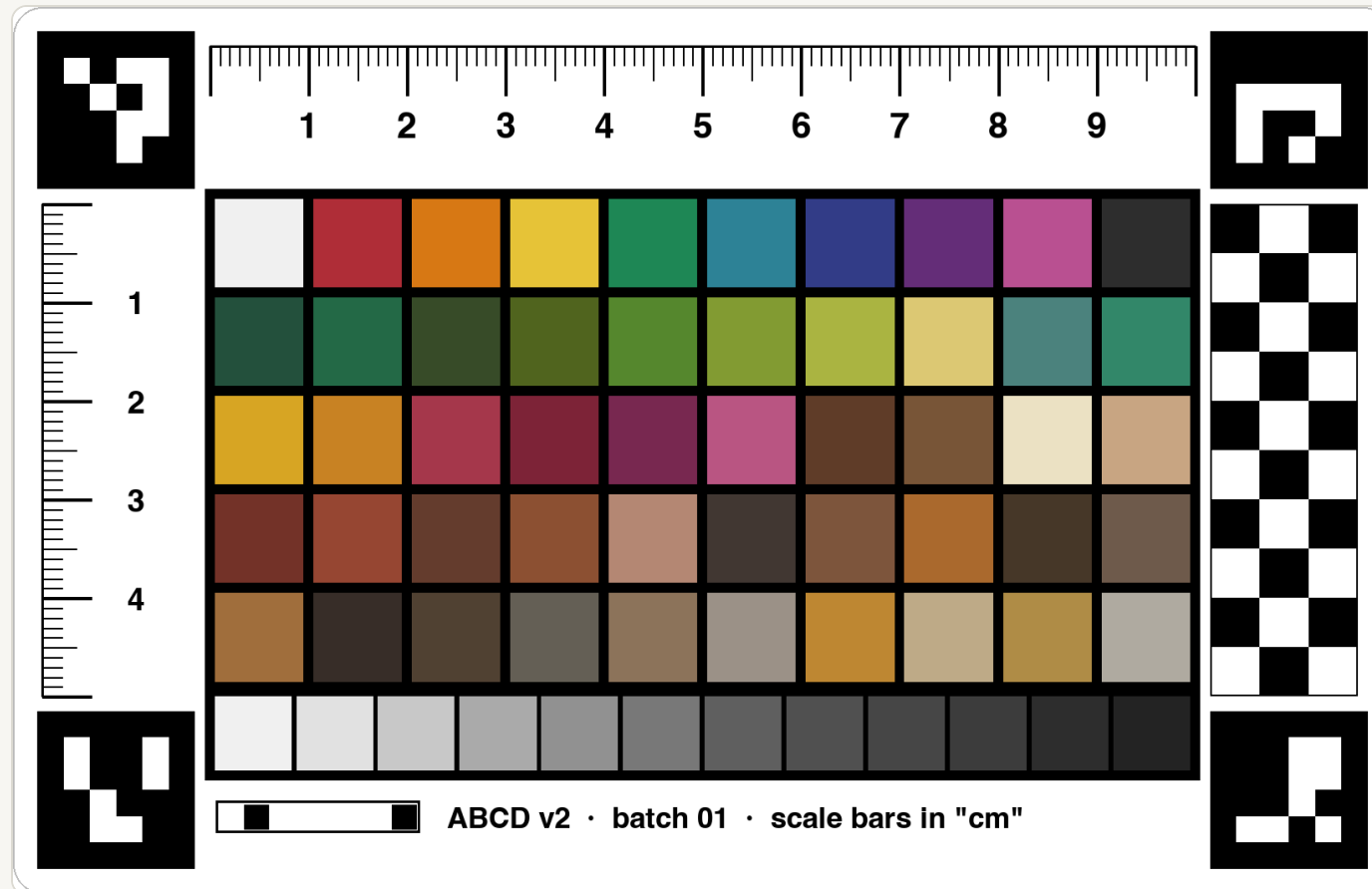
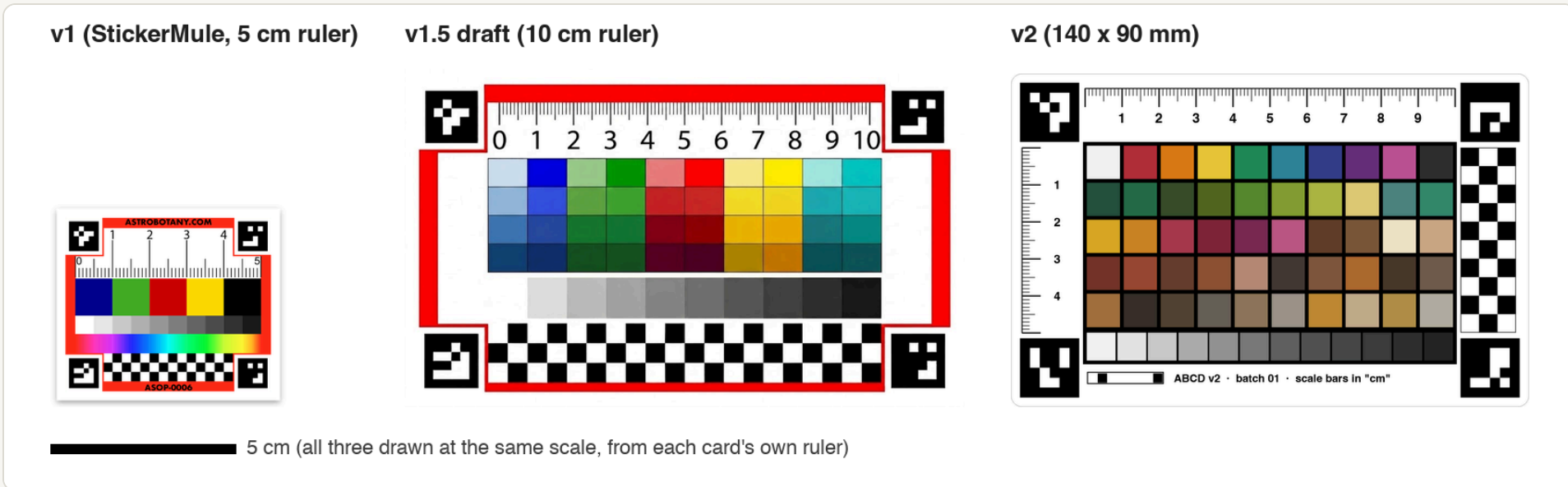


ABCD v2: a bigger calibration card whose colours mean something to plant and soil scientists

140 × 90 mm · 62 Munsell-defined colour chips · self-identifying ArUco corner markers (IDs 0, 1, 2, 3)

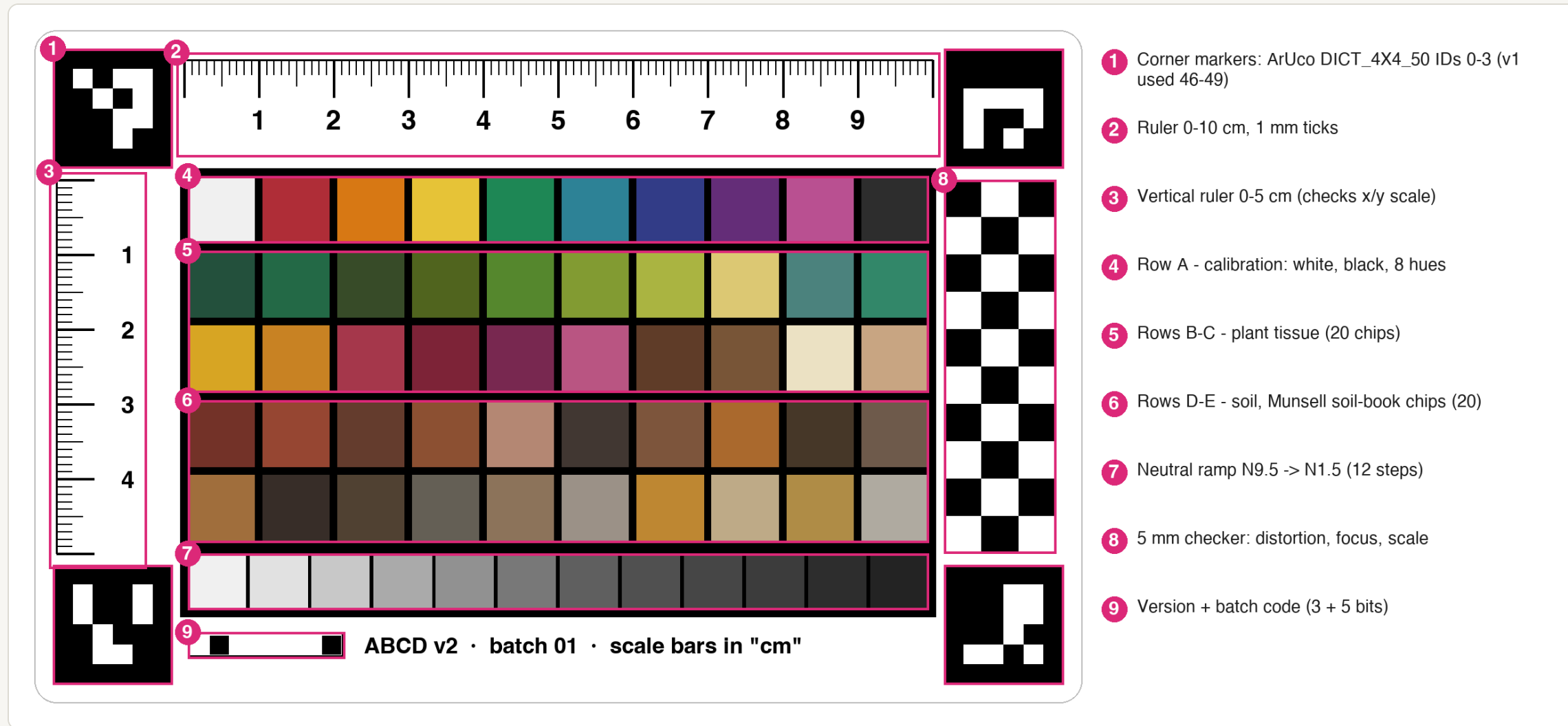


v1 was too small, and its colours didn't map to anything we measure



- **Size:** v1's 5 cm card leaves few pixels per chip; v2 is 140 × 90 mm with 9 mm chips.
- **Colours:** v1/v1.5 used screen primaries. Several can't be printed, and none corresponds to a plant or soil colour class.
- **Red frame:** competes with anthocyanin red in segmentation and pulls auto white balance. Removed.

What is on the card



Every chip is a Munsell notation; the printed colour is computed, then measured

1 • Notation

Soil-chart chips, Plant Tissue book hues, calibration hues.

2 • Renotation

Munsell's published data → CIE xyY.

3 • Print + screen

Lab D50 for print and measuring; sRGB for the file.

4 • Gamut gate

Must print ($\Delta E_{2000} \leq 3$) and fit sRGB, else chroma steps down.

5 • Measure

Spectrophotometer readings per batch replace the design values.

4

chips reduced by the gate: 5Y 8/12 → 5Y 8/10, 2.5G 5/10 → 2.5G 5/8, 5B 5/8 → 5B 5/6, 7.5PB 3/12 → 7.5PB 3/10

2.65

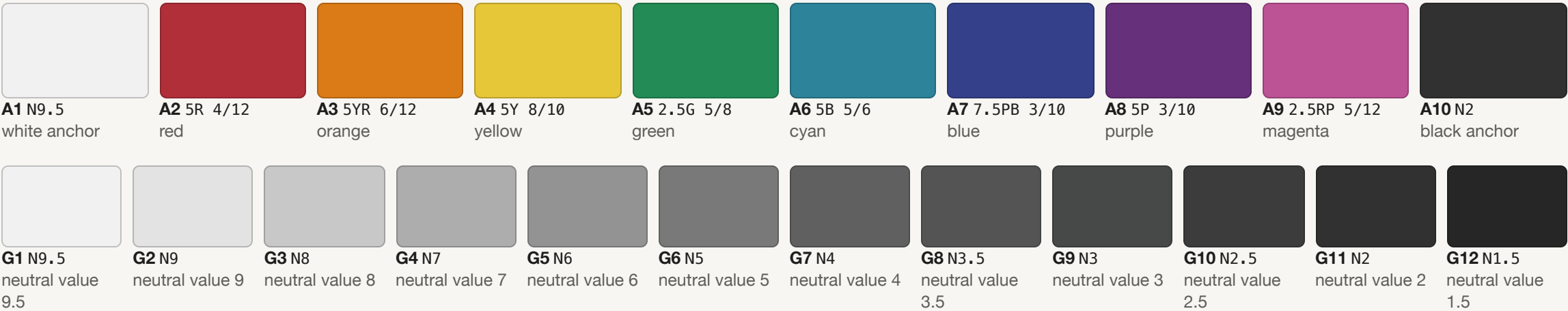
largest remaining ΔE_{2000} between design and printable colour

±3

/255 per channel vs an independent Munsell viewer, 40 chips

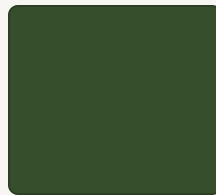
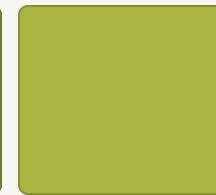
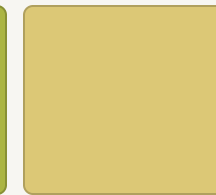
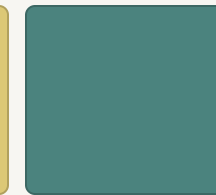
The scanned soil-chart PDF supplied notations and names only; its pixel colours depend on the scanner and were not used.

Calibration: anchors, a hue circle and a 12-step grey ramp



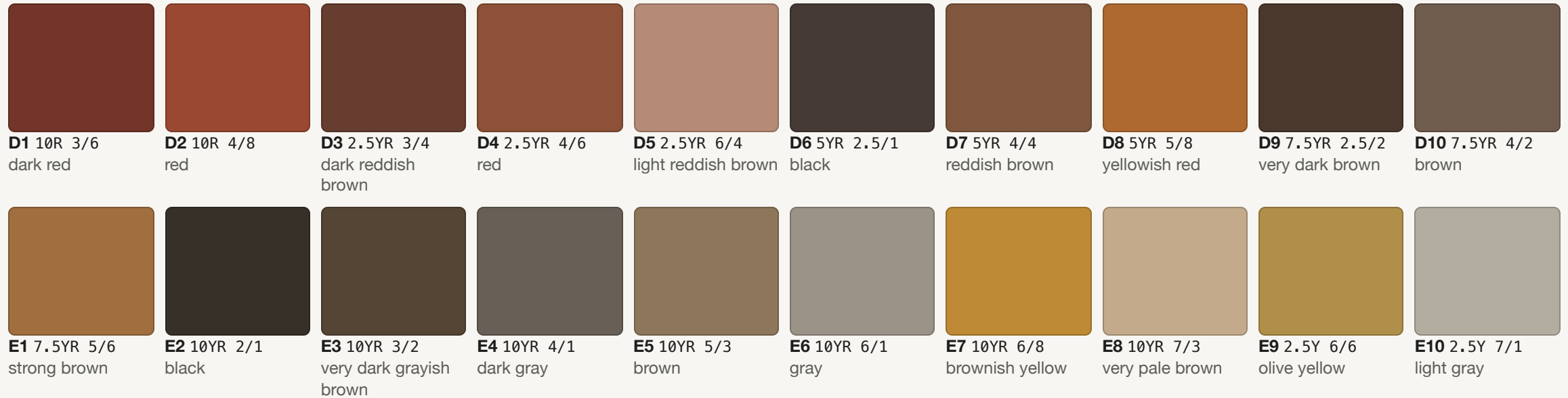
Gives the colour-correction fit something to grip on in every hue direction, plus white, black and N5 (≈ 18 % grey) for exposure and white balance.

Plant tissue: leaf health, pigments, senescence and roots

									
B1 5G 3/4 deep green leaf	B2 2.5G 4/6 dark green leaf	B3 7.5GY 3/4 dark leaf green	B4 5GY 4/6 leaf green	B5 7.5GY 5/8 bright leaf green	B6 5GY 6/8 light green leaf	B7 2.5GY 7/8 chlorotic yellow-green	B8 5Y 8/6 pale chlorotic yellow	B9 5BG 5/4 glaucous blue-green	B10 7.5G 5/6 green stem
									
C1 2.5Y 7/10 senescent yellow	C2 7.5YR 6/10 senescent orange	C3 2.5R 4/10 anthocyanin red	C4 2.5R 3/8 dark anthocyanin red	C5 5RP 3/8 anthocyanin purple-red	C6 5RP 5/10 pink-purple petiole/petal	C7 5YR 3/4 necrotic brown	C8 7.5YR 4/4 browned tissue	C9 2.5Y 9/2 white root / mycelium	C10 7.5YR 7/4 tan root

Every hue is a page of the Munsell Plant Tissue Color Book. **Value/chroma are design targets:** we'd like consultants to confirm or swap them for actual book chips.

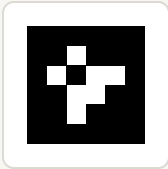
Soil: chips taken from the Munsell soil-chart pages, with their printed names



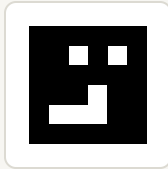
Hues 10R to 2.5Y. The low-chroma greys span grey-regolith darkness. **No chip claims to match a specific simulant or Apollo sample**; targeting simulants needs published Munsell values for them.

Standard ArUco markers: the card identifies itself

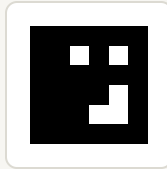
v1 · DICT_4X4_50 IDs 47, 48, 49, 46



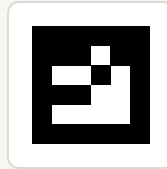
TL · id 47



TR · id 48

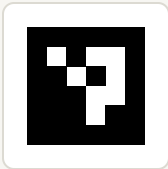


BR · id 49

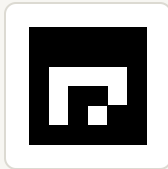


BL · id 46

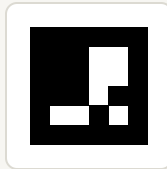
v2 · DICT_4X4_50 IDs 0, 1, 2, 3



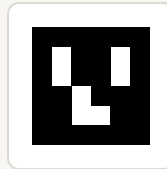
TL · id 0



TR · id 1



BR · id 2



BL · id 3

- v1 already used OpenCV DICT_4X4_50 IDs 46, 47, 48, 49.
- v2 uses IDs 0, 1, 2, 3, the best-separated block of four: at least 5 bits from any v1 marker, 8 from their own rotations.
- Any ArUco reader (OpenCV, arucogen, our app) knows which card, which corner, and which way up.
- Verified with OpenCV's own detector on both cards.

DOES THE APP FIND IT?

Detection on synthetic photos, before and after marker-ID checking

SYNTHETIC PHOTO	V1 · APP BEFORE	V1 · APP AFTER	V2 · APP BEFORE	V2 · APP AFTER
Flat	✓ 1.6 px	✓ 1.7 px · 4/4 IDs · v1	✓ 1.4 px	✓ 1.4 px · 4/4 IDs · v2
Flat + colour cast, noise, blur	✓ 1.4 px	✓ 1.5 px · 4/4 IDs · v1	✓ 1.4 px	✓ 1.4 px · 4/4 IDs · v2
Mild perspective	✓ 1.1 px	✓ 1.1 px · 4/4 IDs · v1	✗ 215.1 px	✓ 1.2 px · 4/4 IDs · v2
Mild perspective + colour cast, noise, blur	✓ 1.1 px	✓ 1.1 px · 4/4 IDs · v1	✗ 133.5 px	✓ 1.2 px · 4/4 IDs · v2
Strong perspective	✓ 2.1 px	✓ 1.3 px · 4/4 IDs · v1	✗ 217.3 px	✓ 1.1 px · 4/4 IDs · v2
Strong perspective + colour cast, noise, blur	✓ 2.0 px	✓ 1.2 px · 4/4 IDs · v1	✗ 220.4 px	✓ 1.2 px · 4/4 IDs · v2
Rotated 15°	✓ 1.3 px	✓ 1.5 px · 4/4 IDs · v1	✓ 1.5 px	✓ 1.5 px · 4/4 IDs · v2
Rotated 15° + colour cast, noise, blur	✓ 1.3 px	✓ 1.3 px · 4/4 IDs · v1	✓ 1.5 px	✓ 1.5 px · 4/4 IDs · v2
Upside down (180°)	✗ 1520.8 px	✓ 1.7 px · 4/4 IDs · v1	✗ 1625.1 px	✓ 0.5 px · 4/4 IDs · v2
Upside down (180°) + colour cast, noise, blur	✗ 1521.1 px	✓ 1.7 px · 4/4 IDs · v1	✗ 1625.1 px	✓ 0.5 px · 4/4 IDs · v2
PASSED	8/10	10/10	4/10	10/10

From design values to measured references

1. Print matte if possible.
2. Measure ≥ 3 stickers per batch, every chip, with a spectrophotometer (D50 / 2°).
3. Flag chips that vary by more than $\Delta E_{2000} 2$ between stickers.
4. Measured Lab values replace design values in the app.
5. Each print run gets its own batch number on the card.

v1 scale finding. Against its own printed ruler, v1's markers are not 4.3 cm apart in both directions as the app assumes:

ruler: 94.3 px/cm (0 cm at x=63.0, 5 cm at x=534.5)
fiducial-centre span: x 4.56 cm, y 3.69 cm (mean 4.12 cm)
app assumes 4.30 cm for both -> scale error -4.1% (lengths reported +4.3%)

To confirm on real photos before changing the app.

Open questions for consultants

Plant shades. Right tissue classes and shades? Swap for real Plant Tissue book chips?

Soil targets. Should chips match specific regolith simulants? Which, with what published values?

Format. Sticker or laminated card? Is 14 × 9 cm right for the imaging setups we support?

Markers + numbering. IDs 0–3 for v2; how should future versions and batches be numbered?

v1 scale. Confirm the 4.3 cm discrepancy on real v1 photos, then correct the app.

Measurement. Who measures printed batches, and with which instrument?

Comment on the GitHub issues: github.com/dr-richard-barker/ABCD_v2/issues